ADITYA RAJ

Pre-Final Year B.Tech — Electronics & Communication Engineering

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Professional Summary

Pre-final year Electronics & Communication Engineering student with practical experience in Embedded Systems, UAV Electronics, Flight Controllers and AI/ML. Former Research Intern at the Drone Electronics Lab, IIT (ISM) Dhanbad with hands-on experience in Pixhawk integration, STM32 firmware, sensor interfacing, autonomous UAV systems and intelligent embedded applications.

Education

Birla Institute of Technology, Mesra 2024–Present
B.Tech in Electronics & Communication Engineering CGPA: 8.30

Experience

Research Intern — Drone Electronics Lab, IIT (ISM) Dhanbad May 2026–Jul 2026

- Developed a custom STM32 Black Pill flight controller integrating MPU6050, BMP280, NEO-6M GPS and HMC5883L.
- Configured Pixhawk, ArduPilot, Mission Planner, and MAVLink for quadcopter and hexacopter platforms.
- Performed sensor integration, ESC calibration, hardware validation, and successful flight testing of UAV systems.
- Assisted in UAV assembly, system debugging, and embedded hardware integration.

AI & Data Analytics Intern — Shell India – Edunet Foundation Jul 2025–Aug 2025

- Developed an Employee Salary Predictor using Random Forest Regression and Streamlit.
- Performed data preprocessing, feature engineering, visualization and model evaluation.

Projects

Custom STM32 Flight Controller

- Designed and developed a custom UAV flight controller using STM32 Black Pill.
- Integrated MPU6050, BMP280, NEO-6M GPS and HMC5883L using UART and I^2C communication.
- Developed embedded firmware for sensor acquisition and hardware testing.

Gesture Controlled Drone

- Developed a wearable gesture transmitter using ESP32 and MPU6050.
- Implemented ESP-NOW for low-latency wireless communication.
- Used MAVLink to interface with Pixhawk for drone control.

Air Mouse using ESP32 & MPU6050

- Developed an ESP32 Bluetooth HID Air Mouse using MPU6050 for real-time gesture-based cursor control.

Employee Salary Predictor

- Machine learning web application using Python, Random Forest Regression and Streamlit.

Technical Skills

Programming
Python, C, C++, HTML, CSS

Embedded
STM32, ESP32, ESP8266, Arduino UNO/Nano, Raspberry Pi

Drone Technologies
Pixhawk, ArduPilot, Mission Planner

Protocols
UART, I2C, SPI, PWM, Bluetooth, MAVLink, ESP-NOW

Sensors
MPU6050, BMP280, NEO-6M GPS, HMC5883L, DHT11, MQ2, HC-SR04, RFID

AI / ML
Scikit-learn, Pandas, NumPy, Matplotlib, Streamlit

Tools
GitHub, Arduino IDE, VS Code, Jupyter Notebook

Certifications

- Python for AI, ML & DL – IIT Kanpur
- Research Internship – IIT (ISM) Dhanbad
- AI with Data Analytics – Shell India
- IBM Data Fundamentals
- Python Programming – Reliance Foundation
- Verilog Internship – NIELIT

Technical Interests

Embedded Systems • UAV & Drone Electronics • IoT • AI/ML for Embedded Systems